①a $X_t = W_t^2$ $\quad dX_t$

$$X_t = X_0 + \int_0^t F_s ds + \int_0^t G_s dW_s \iff dX_t = F_t dt + G_t dW_t$$

$$u(t,x)$$

$$\boxed{u(t,X_t) = u(0,X_0) + \int_0^t u_x(s,X_s)G_s dW_s + \int_0^t \{u'(s,X_s) + u_x(s,X_s)F_s + \tfrac{1}{2}u_{xx}(s,X_s)G_s^2\} ds}$$

$$du(t,X_t) = u'(t,X_t)dt + \partial u(t,X_t) dX_t + \tfrac{1}{2}\partial^2 u(t,X_t)(dX_t)^2$$

$$\begin{array}{c|cc} & dt & dW \\ \hline dt & 0 & 0 \\ dW & 0 & \sigma^2 dt \end{array} \qquad \sigma = 1$$

$u(t,x) = x^2$ $\quad u_x(t,x) = 2x$

$u'(t,x) = 0$ $\quad u_{xx}(t,x) = 2$

$$dX_t = d(W_t^2) = du(t,W_t) = 2W_t dW_t + \tfrac{1}{2}\cdot 2\cdot(dW_t)^2 = \underline{2W_t dW_t + dt}$$

①b $X_t = W_t^4$ $\quad dX_t$

$u(t,x) = x^4$ $\quad u_x(t,x) = 4x^3$

$u_t(t,x) = 0$ $\quad u_{xx}(t,x) = 12x^2$

$$dX_t = d(W_t^4) = du(t,W_t) = 4W_t^3 dW_t + 6W_t^2 dt$$

② $X_t = e^{W_t}$

$u(t,x) = e^x$

$u' = 0$

$u_x = e^x$

$u_{xx} = e^x$

$$dX_t = du(t,W_t) = e^{W_t}dW_t + \tfrac{1}{2}e^{W_t}(dW_t)^2 =$$

$$\underline{dX_t = X_t dW_t + \tfrac{1}{2}X_t dt}$$

стохастическое дифферен. уравнение (СДУ)

③ $\int_0^t \sin W_s dW_s$ — вычислить бу через интеграл Ито.

$$u(t,X_t) = u(0,X_0) + \int_0^t \sin W_s dW_s + \int_0^t (u'(s,X_s) + u_x(s,X_s)\cancel{F_s} + \tfrac{1}{2}u_{xx}(s,X_s)G_s^2\} ds$$

$G_t = 1$ $\quad F_s = 0$ $\quad G_s^2 = 1$

$$\int_0^t u_x(s,X_s) G_s dW_s \qquad G_s = 1$$

$u_x(s,x) = \sin x$

$X_t = X_0 + \cancel{\int_0^t F_s ds} + \int_0^t G_t dW_t$

$\underline{dX_t = dW_t + F_t dt}$

$u_x(s,W_s) = \sin W_s \Rightarrow u(s,W_s) = -\cos W_s$

$X_s = W_s$

$u(t,x) = -\cos x$

$u_{xx}(t,x) = \cos x$ $\quad u'(t,x) = 0$

$$-\cos W_t = -\cos W_0 + \int_0^t \sin W_t dW_t + \tfrac{1}{2}\int \cos W_t ds$$

$$\underline{\int_0^t \sin W_t dW_t} = 1 - \cos W_t - \tfrac{1}{2}\int_0^t \cos W_t ds$$

$$= 1 - \cos W_t - \tfrac{1}{2}\int_0^t \cos W_t ds \cancel{- \int_0^t \sin W_s ds}$$

④ $X_t = e^{\sigma W_t + \mu t}$ $\quad dX_t$ $\qquad dX_t = f_1(t,X_t)dt + f_2(t,X_t)dW_t$

$u(t,x) = e^{\sigma x + \mu t}$

$u'(t,x) = \mu e^{\sigma x + \mu t}$

$u_x(t,x) = \sigma e^{\sigma x + \mu t}$

$u_{xx}(t,x) = \sigma^2 e^{\sigma x + \mu t}$

$$dX_t = \mu \underbrace{e^{\sigma W_t + \mu t}}_{X_t} dt + \sigma \underbrace{e^{\sigma W_t + \mu t}}_{X_t} dW_t + \tfrac{1}{2}\sigma^2 \underbrace{e^{\sigma W_t + \mu t}}_{X_t} \overset{(dW_t)^2}{dt} =$$

$$= \mu X_t dt + \sigma X_t dW_t + \tfrac{1}{2}\sigma^2 X_t dt =$$

$$\underline{dX_t = (\mu + \tfrac{1}{2}\sigma^2)X_t dt + \sigma X_t dW_t}$$

⑤ $dY_t = 0.4\, dW_t + 0.1\, dt$ $\qquad \underline{X_t = e^{0.5 Y_t}}$

dX_t ?

$$dX_t = \frac{e^{0.5 Y_t}}{2} dY_t + \frac{e^{0.5 Y_t}}{4\cdot 2}(dY_t)^2 = \frac{X_t}{2}\cdot\left(\frac{4}{10}dW_t + \frac{1}{10}dt\right) + \frac{X_t}{4\cdot 2}\left(\frac{4}{10}dW_t + \frac{1}{10}dt\right)^2 =$$

$u(t,x) = e^{0.5x}$

$$= \frac{X_t}{5}dW_t + \frac{X_t}{20}dt + \frac{X_t}{50}dt = \underline{\frac{X_t}{5}dW_t + \frac{7X_t}{100}dt}$$

$u' = 0$

$u_x = \dfrac{e^{0.5x}}{2}$ $\qquad u_{xx} = \dfrac{e^{0.5x}}{4}$

⑥ $L_t = e^{X_t - \frac{1}{2}[X]_t}$ $\qquad L_t = 1 + \int_0^t L_s dX_s$

$Y_t = [X]_t$

X_t

$u(x,y) = e^{x - \frac{y}{2}}$

$$du(X_t,Y_t) = u_x dX_t + u_y dY_t + \tfrac{1}{2}u_{xx}(dX_t)^2 + \cancel{\tfrac{1}{2}u_{yy}\overset{0}{(dY_t)^2}} + \cancel{u_{xy}dX_t dY_t} =$$

$u_x = e^{x - \frac{y}{2}}$ $\qquad u_{yy} = \tfrac{1}{4}e^{x - \frac{y}{2}}$

$u_y = -\tfrac{1}{2}e^{x - \frac{y}{2}}$ $\qquad u_{xy} = -\tfrac{1}{2}e^{x - \frac{1}{2}y}$

$u_{xx} = e^{x - \frac{y}{2}}$

$$(dX_t)^2 = d[X] = dY_t$$

$$L_t = u(X_t, Y_t)$$

$$(dY_t)^2 = 0$$

$$dX_t dY_t = 0$$

$$= L_t dX_t - \tfrac{1}{2}L_t dY_t + \tfrac{1}{2}L_t dY_t =$$

$$= L_t dX_t$$

$$dL_t = L_t dX_t \quad \Big| \int \qquad L_0 = e^0 = 1$$

$$L_t = L_0 + \int_0^t L_s dX_s$$

$$L_t = 1 + \int_0^t L_s dX_s$$